Neo-Hookean Liver Deformation: Model, Numerical Solution, and Implementation Outline

18 September 2026

Scope. This document describes the deformation problem implemented in `simulate_deformation/`, including its current Elmer interface. It then outlines a custom solver for the same boundary-value problem. The custom solver and incremental-loading procedure are proposed designs, not implemented features of the current generation script. No voxelization or registration is required to generate a deformation.

Example outline

1. Input mesh, units, and unknown displacement field.
2. Fixed boundary and traction-patch sampling.
3. Neo-Hookean material and equilibrium equations.
4. Tetrahedral discretization and force assembly.
5. Nonlinear updates, linear solves, and line search.
6. Convergence criteria, failure classification, and output validation.
7. Current file workflow and reproducible settings.
8. Implementation outline for a custom solver.

1 Input and unknowns

We start with a volumetric liver mesh consisting of N vertices $\boldsymbol{X}_a \in \mathbb{R}^3$ and four-node tetrahedra. Input coordinates to `main.py` are in metres. Young's modulus is in pascals, traction is in pascals, and forces are in newtons. The MSD batch wrapper converts source coordinates from millimetres to metres before invoking this script.

The unknown is the displacement $\boldsymbol{u}_a$ at every vertex. The deformed position is $\boldsymbol{x}_a = \boldsymbol{X}_a + \boldsymbol{u}_a$. We solve a static equilibrium problem without inertia or body forces. The reference configuration is Ω_0; its boundary is divided into a fixed region Γ_D, loaded patches $\Gamma_{p,0}$, and a traction-free remainder.

2 Boundary conditions and sampling

Exterior triangles are extracted by identifying tetrahedral faces that occur only once. Patches grow through triangle-edge adjacency from a randomly selected surface vertex and an incident triangle, until their accumulated reference area reaches the requested target. These are connected surface patches, not geometric circles. The area overshoot is less than the area of the final added triangle.

For independent draws $U \sim \mathcal{U}[0,1)$, the current sampler uses

$$r_D = 0.025\,\mathrm{m} + 0.05\,U L_{\max}, \qquad A_D^{\mathrm{target}} = 3\pi r_D^2, \tag{1}$$

$$r_p = (0.01 + 0.06U)\,\mathrm{m}, \qquad A_p^{\mathrm{target}} = \pi r_p^2. \tag{2}$$

Here $L_{\max}$ is the largest axis-aligned extent of the reference mesh. The first term in r_D is an absolute length; only its second term scales with liver size. The force-patch radius parameter ranges from 1 to 7 cm. The single-sample default samples one to three force patches; the batch default samples two to three.

Force patches exclude the fixed patch and a conservative 2 cm Euclidean clearance around it. A triangle is eligible only if its centroid-to-fixed-surface distance minus its maximum centroid-to-vertex distance is at least 2 cm. Force patches cannot share triangles, although they may share edges or vertices. An insufficient eligible connected area produces a preparation failure without resampling, shrinking a patch, or reducing a force.

All three displacement components are zero on the fixed patch. For each loaded patch, we sample a nominal force magnitude and direction as

$$q_p = s(1 - U_p^2)\,\mathrm{N}, \qquad \boldsymbol{d}_p = \frac{\boldsymbol{v}_p}{\|\boldsymbol{v}_p\|_2}, \qquad (v_p)_i \sim \mathcal{U}[-0.5, 0.5). \tag{3}$$

The direction distribution is induced by normalizing a vector sampled in a cube; it is not exactly uniform on the sphere. The force-factor parameter s is an upper bound on each nominal force draw, not the force assigned to every patch. Using the actual grown reference area $A_{p,0}$, the prescribed Cartesian traction is

$$\boldsymbol{t}_p = \frac{q_p \boldsymbol{d}_p}{A_{p,0}}. \tag{4}$$

Its components are rounded to six decimal places before writing the SIF.

Area convention in the current Elmer case. The SIF specifies `Force 1`, `Force 2`, and `Force 3`, and does not enable `Pseudo-Traction`. In the installed Elmer implementation, these are Cartesian tractions per unit *deformed* surface area. Consequently, the prescribed components remain constant but the total patch force varies with its deformed area. Equation (4) normalizes the force at the reference configuration. A constant force obtained by integrating over reference area would be a different loading convention and must be selected explicitly in a replacement.

3 Material and equilibrium

We model the liver as homogeneous, isotropic, compressible neo-Hookean material. Young's modulus is specified by the caller or sampled uniformly from 500 to 5000 Pa. The default Poisson ratio is $\nu = 0.35$. The Lamé parameters are

$$\mu = \frac{E}{2(1+\nu)}, \qquad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}. \tag{5}$$

Define the deformation gradient, volume ratio, and right Cauchy–Green tensor by

$$\boldsymbol{F} = \boldsymbol{I} + \nabla_X \boldsymbol{u}, \qquad J = \det \boldsymbol{F} > 0, \qquad \boldsymbol{C} = \boldsymbol{F}^T \boldsymbol{F}. \tag{6}$$

The strain-energy density corresponding to the stress law in Elmer's `NeoHookeanLocalMatrix` is

$$W(\boldsymbol{F}) = \frac{\mu}{2}(\operatorname{tr} \boldsymbol{C} - 3) - \mu \ln J + \frac{\lambda}{4}(J^2 - 1 - 2\ln J). \tag{7}$$

Its first Piola–Kirchhoff stress is

$$\boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}} = \mu(\boldsymbol{F} - \boldsymbol{F}^{-T}) + \frac{\lambda}{2}(J^2 - 1)\boldsymbol{F}^{-T}. \tag{8}$$

Several distinct compressible material laws are called neo-Hookean. Replacing Equation (7) with a different volumetric energy can change the deformation even when E and ν are unchanged.

Equilibrium requires zero force imbalance for all admissible virtual displacements $\boldsymbol{w}$ that vanish on Γ_D:

$$\int_{\Omega_0} \boldsymbol{P} : \nabla_X \boldsymbol{w}\, dV_0 - \sum_p \int_{\Gamma_{p,0}} \boldsymbol{w} \cdot \boldsymbol{t}_p\, j_s\, dA_0 = 0, \qquad j_s = J\|\boldsymbol{F}^{-T}\boldsymbol{N}\|_2. \tag{9}$$

Here $\boldsymbol{N}$ is the reference surface normal and j_s is the deformed-to-reference area ratio. Setting $j_s = 1$ in the load integral instead describes prescribed reference-area traction. The unforced boundary has zero prescribed traction but is free to move.

4 Tetrahedral discretization

Within each tetrahedron e, linear shape functions N_a interpolate displacement:

$$\boldsymbol{u}_h(\boldsymbol{X}) = \sum_{a=1}^{4} N_a(\boldsymbol{X})\boldsymbol{u}_a, \qquad \boldsymbol{F}_e = \boldsymbol{I} + \sum_{a=1}^{4} \boldsymbol{u}_a \otimes \nabla_X N_a. \tag{10}$$

The shape-function gradients and reference volume $V_{e,0}$ are constant in a linear tetrahedron and can be precomputed. The element's internal nodal force is

$$\boldsymbol{f}_{e,a}^{\text{int}} = V_{e,0}\boldsymbol{P}_e \nabla_X N_a. \tag{11}$$

For uniform traction on a boundary triangle of deformed area $A_f(\boldsymbol{u})$, each of its three vertices receives the elemental contribution

$$\boldsymbol{f}_{f,a}^{\text{ext}} = \frac{A_f(\boldsymbol{u})}{3}\boldsymbol{t}_p. \tag{12}$$

Contributions are accumulated at shared vertices. Boundary group IDs select the prescribed traction; the native boundary file itself stores no force values.

Assembly gives the global residual

$$\boldsymbol{r}(\boldsymbol{u}) = \boldsymbol{f}^{\text{int}}(\boldsymbol{u}) - \boldsymbol{f}^{\text{ext}}(\boldsymbol{u}). \tag{13}$$

The equilibrium equations are solved on unconstrained degrees of freedom. Nonzero residuals at fixed vertices represent support reactions and must not be treated as unbalanced free-node forces.

5 Nonlinear solution and line search

A Newton formulation computes an update from

$$\boldsymbol{K}_k \, \Delta \boldsymbol{u}_k = -\boldsymbol{r}(\boldsymbol{u}_k), \qquad \boldsymbol{K}_k = \left. \frac{\partial \boldsymbol{r}}{\partial \boldsymbol{u}} \right|_{\boldsymbol{u}_k}, \qquad \boldsymbol{u}_{k+1} = \boldsymbol{u}_k + \alpha \Delta \boldsymbol{u}_k. \tag{14}$$

The tangent changes because material response and geometry depend on the current displacements. For the current-area traction convention, a fully consistent tangent also contains the derivative of the external nodal load. Elmer's boundary routine states that these load dependencies are evaluated using the previous iterate; an independently implemented consistent Newton tangent therefore need not reproduce Elmer's iteration history, even for the same equilibrium problem.

The current default solves each linearized system using CG with Cholesky2 preconditioning, a maximum of 500 linear iterations, and a linear tolerance of 10^{-7}. CG requires a symmetric positive-definite system; this property is not guaranteed for every large-deformation tangent. GMRES/ILU2 and BiCGStabl/ILU2 templates are also available. Replacing the linear solver does not change the intended equilibrium equations.

Elmer's default line-search cost is the norm of the assembled algebraic residual. Writing this cost as c, the configured search accepts a trial when

$$c(\boldsymbol{u}_k + \alpha \Delta \boldsymbol{u}_k) \leq (1 - \eta\alpha) c(\boldsymbol{u}_k). \tag{15}$$

The full step starts at $\alpha = 1$ and rejected steps are halved. The current templates inherit Elmer's defaults of $\eta = 0.5$ and ten trials. A separate replay experiment used $\eta = 0.1$, requiring a 10% cost reduction for a full step instead of 50%. A trial reassembles and evaluates the residual at a new displacement; it does not perform another CG solve. The same update direction is reused until the trial is accepted or the search fails.

6 Stopping and validation

The current line-search path uses the change in the scalar displacement norm as its nonlinear stopping measure. For M displacement components,

$$n_k = \sqrt{\frac{1}{M} \sum_{i=1}^{M} u_{k,i}^2}, \tag{16}$$

$$\delta_k = \frac{2|n_k - n_{k-1}|}{n_k + n_{k-1}} < 10^{-4}. \tag{17}$$

The zero-denominator case is handled separately by Elmer. This dimensionless threshold corresponds approximately to a change below 0.01% in the overall RMS displacement. It is neither a per-vertex displacement threshold nor a force equilibrium tolerance. In particular, the difference between two norms is not the norm of the difference between two displacement fields.

The script accepts a simulation only when the solver exits successfully, reports no Elmer error, produces its result file, and has a finite final logged nonlinear change below the SIF tolerance. It additionally requires finite coordinates and displacements and no tetrahedron with zero or reversed

signed volume relative to its reference state. These checks do not independently establish a specified final force-residual tolerance or exclude surface self-intersection.

Failure category	Interpretation
Preparation failure	An eligible connected surface region cannot supply the sampled patch area.
Line-search failure	None of the allowed trial step lengths satisfies sufficient reduction.
Linear divergence	The linear residual exceeds the solver's divergence limit; this is distinct from reaching the iteration cap.
Nonlinear nonconvergence	The final logged change fails the required tolerance, commonly after exhausting 100 iterations.
Invalid geometry	Nonfinite output or a collapsed/inverted tetrahedron is detected.

7 Current implementation and files

The input VTK is read into coordinate and connectivity arrays. These directly generate the following Elmer input files:

File	Content
`mesh.nodes`	Node IDs and reference coordinates.
`mesh.elements`	Tetrahedral connectivity and body IDs.
`mesh.header`	Counts and element types.
`mesh.boundary`	Surface triangles, parent tetrahedra, and boundary group IDs.
`case.sif`	Material properties, group boundary values, and solver options.

Group 1 is free, group 2 is fixed, and groups 3 onward represent individual load patches. Elmer reads these native files; it does not read the source VTK directly. There are no intermediate `volume.vtk` or `simulation.vtk` files.

The default template specifies one steady-state solve, 100 nonlinear iterations, and enabled line search. It also records Newton-after-iterations 50, Newton-after-tolerance 10^{-3}, and relaxation factor 1. These keywords should not be interpreted as proof of 50 initial Picard steps: the installed neo-Hookean element routine explicitly constructs a Newton material tangent. No load ramp is currently configured.

`manifest.json` records the arguments, sampled material, patch seeds and areas, prescribed tractions, full SIF, timing, and status. Elmer writes `result/case_t0001.vtu`; its coordinates are already deformed. On acceptance, the script writes `deformed.vtk`. Displacements must not be added to those deformed coordinates a second time. Batch processing additionally records per-sample metrics and optional rigid-registration TRE.

For example, an input already expressed in metres can be processed by:

```
pixi run --manifest-path baselines/Volume2SurfaceCNN/pixi.toml python \
  simulate_deformation/main.py liver.vtk outputs/example_deformation \
  --seed 1 --force-factor 3 --poisson-ratio 0.35 \
  --youngs-modulus 3000 --min-force-patches 2 --template cg
```

The output directory must be new. The line-search replay changed only the following SIF setting; it did not modify the default templates:

```
Nonlinear System Linesearch Limit = 0.1
```

8 Example custom-solver outline

A replacement can retain the existing mesh reader, surface extraction, patch sampler, manifests, and VTK output. It needs a new material/element kernel, sparse assembly, and nonlinear driver. Existing sparse linear algebra libraries can solve the linear systems; implementing CG or a sparse factorization from scratch is unnecessary.

```
read reference vertices and tetrahedra in metres
extract surface; sample fixed and traction patches
precompute tetrahedral gradients, volumes, and assembly indices
identify fixed and free displacement components
u = 0

for load_fraction in [0.1, 0.2, ..., 1.0]:
    scale prescribed tractions by load_fraction
    for nonlinear_iteration in range(max_iterations):
        compute element F, det(F), stresses, and internal forces
        assemble external forces using the chosen area convention
        assemble residual on free components
        check explicitly defined residual convergence
        assemble consistent tangent; solve K * du = -residual

        alpha = 1
        for trial in range(max_line_search_trials):
            trial_u = u + alpha * du
            reject trial if any tetrahedron has det(F) <= 0
            recompute trial residual
            if sufficient reduction is satisfied:
                accept trial_u; break
            alpha *= 0.5
        record residual, update size, alpha, and iteration counts
        stop and record failure if no trial is accepted
    stop if the load increment did not converge

validate geometry and final force residual at load_fraction = 1
write deformed VTK and manifest
```

This proposed continuation procedure retains the same mesh, material, patches, and final tractions. A partial-load solution must not be reported as a successful full-load deformation. Incremental loading is also discussed for the FEM data generation in U-Mesh [2].

For an independent solver, we would explicitly monitor the free-node residual, for example $\|\boldsymbol{r}_{\mathrm{free}}\|_2/\|\boldsymbol{f}^{\mathrm{ext}}_{\mathrm{free}}\|_2$ when the denominator is nonzero, together with displacement-update size and element Jacobians. Its tolerance would need validation; it must not be assigned 10^{-4} on the assumption that it is equivalent to Equation (17).

Verification should include zero-load and rigid-motion material checks, finite-difference checks of element forces and tangent derivatives, a small-deformation comparison against linear elasticity, and matched comparisons with converged Elmer cases. Matching the energy, traction convention, and constraints is necessary before attributing differences to solver accuracy.

Implementation references

The implementation described here is based on the repository state inspected on 18 September 2026. The main source locations are:

- `simulate_deformation/boundaries.py`: patch and force sampling.
- `simulate_deformation/main.py`: preparation, execution, and acceptance.
- `simulate_deformation/case.sif.template`: default solver settings.
- `simulate_deformation/elmer_mesh.py`: native mesh writing.
- `simulate_deformation/mesh_utils.py`: geometry validation.
- Elmer `fem/src/modules/ElasticSolve.F90`: `NeoHookeanLocalMatrix` and `LocalBoundaryMatrix`.
- Elmer `fem/src/SolverUtils.F90`: `CheckStepSize` and `ComputeNorm`.

The inspected Elmer installation reports version 26.2, revision `a19504a`; its source is stored under `baselines/Volume2SurfaceCNN/.elmer-source/`. Its formulations are documented in the Elmer Models Manual [1]. LIBR's `baselines/LIBR/src/libr/fem.py` supplies reusable linear-tetrahedron geometry and assembly patterns, but its linear material stiffness is not a neo-Hookean solver.

References

[1] CSC–IT Center for Science. *Elmer Models Manual*, chapter on finite elasticity. `https://www.nic.funet.fi/index/elmer/doc/ElmerModelsManual.pdf`. The installed source is the authority for the implementation details above.

[2] A. Mendizabal, P. Márquez-Neila, and S. Cotin. Simulation of hyperelastic materials in real-time using deep learning. *Medical Image Analysis*, 59:101569, 2020. `https://arxiv.org/abs/1904.06197`.